Lab 7: Fuzzing for Input Validation Bugs

NAME:ANANYA SURESH

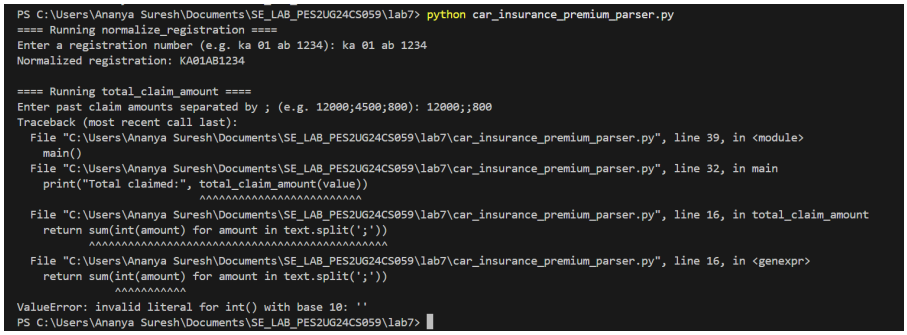
SRN:PES2UG24CS059

CLASS:5A

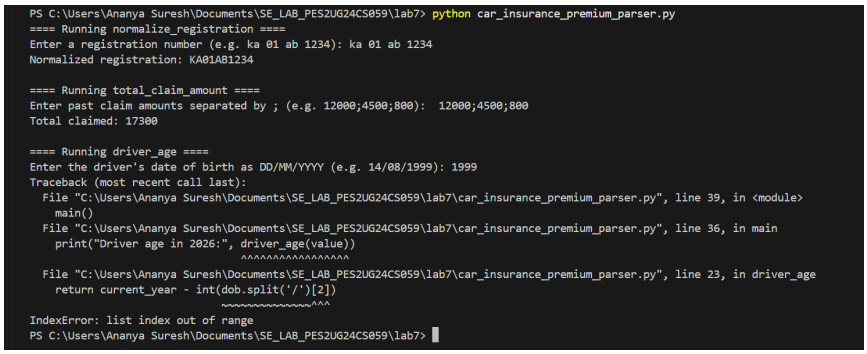
Buggy module run

`python car_insurance_premium_parser.py` on the original module. The claims input `12000;;800` crashes with a `ValueError`, and the date of birth `1999` crashes with an `IndexError`.

claims crash, `12000;;800`:



date of birth crash, `1999`:



Tests failing on the buggy module

`python -m pytest test_car_insurance_premium_parser.py` on the original module. All 3 Hypothesis tests fail, because each function crashes on `None`, and the claims and date functions also crash on malformed strings.

```
PS C:\Users\Ananya Suresh\Documents\L6_L7_55_car_insurance_premium\LAB7>     python -m pytest test_car_insurance_premium_parser.py --tb=short
============================================================ test session starts ============================================================
platform win32 -- Python 3.12.6, pytest-8.3.4, pluggy-1.5.0
rootdir: C:\Users\Ananya Suresh\Documents\L6_L7_55_car_insurance_premium\LAB7
plugins: anyio-4.15.1, hypothesis-6.170.14
collected 3 items

test_car_insurance_premium_parser.py FFF                                                                                           [100%]

================================================================= FAILURES ==================================================================
_____________________________________ test_normalize_registration_no_crash ______________________________________
test_car_insurance_premium_parser.py:8: in test_normalize_registration_no_crash
    normalize_registration(s)
car_insurance_premium_parser.py:6: in normalize_registration
    plate = plate.strip().upper()
E   AttributeError: 'NoneType' object has no attribute 'strip'

During handling of the above exception, another exception occurred:
test_car_insurance_premium_parser.py:6: in test_normalize_registration_no_crash
    def test_normalize_registration_no_crash(s):
test_car_insurance_premium_parser.py:10: in test_normalize_registration_no_crash
    assert False, f"normalize_registration crashed with: {e!r}"
E   AssertionError: normalize_registration crashed with: AttributeError("'NoneType' object has no attribute 'strip'")
E   assert False
E   Failing test case: test_normalize_registration_no_crash(
E       s=None,
E   )
E   Explanation:
E       These lines were always and only run by failing test cases:
E           C:\Users\Ananya Suresh\Documents\L6_L7_55_car_insurance_premium\LAB7\test_car_insurance_premium_parser.py:9
_____________________________________ test_total_claim_amount_no_crash ______________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\runner.py", line 341, in from_call
  |     result: TResult | None = func()
  |                             ^^^^^^
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\runner.py", line 242, in <lambda>
  |     lambda: runtest_hook(item=item, **kwds), when=when, reraise=reraise
  |             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_hooks.py", line 513, in __call__
  |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_manager.py", line 120, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_callers.py", line 182, in _multicall
  |     return outcome.get_result()
  |            ^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_result.py", line 100, in get_result
  |     raise exc.with_traceback(exc.__traceback__)
```

```
File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_callers.py", line 167, in _multicall
  teardown.throw(outcome._exception)
File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\threadexception.py", line 92, in pytest_runtest_call
  yield from thread_exception_runtest_hook()
File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\threadexception.py", line 68, in thread_exception_runtest_hook
  yield
File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_callers.py", line 167, in _multicall
  teardown.throw(outcome._exception)
File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\unraisableexception.py", line 95, in pytest_runtest_call
  yield from unraisable_exception_runtest_hook()
File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\unraisableexception.py", line 70, in unraisable_exception_runtest_hook
  yield
File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_callers.py", line 167, in _multicall
  teardown.throw(outcome._exception)
File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\logging.py", line 846, in pytest_runtest_call
  yield from self._runtest_for(item, "call")
File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\logging.py", line 829, in _runtest_for
  yield
File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_callers.py", line 167, in _multicall
  teardown.throw(outcome._exception)
File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\capture.py", line 880, in pytest_runtest_call
  return (yield)
          ^^^^^
File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_callers.py", line 167, in _multicall
  teardown.throw(outcome._exception)
File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\skipping.py", line 257, in pytest_runtest_call
  return (yield)
          ^^^^^
File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_callers.py", line 103, in _multicall
  res = hook_impl.function(*args)
        ^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\runner.py", line 174, in pytest_runtest_call
  item.runtest()
File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\python.py", line 1627, in runtest
  self.ihook.pytest_pyfunc_call(pyfuncitem=self)
File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_hooks.py", line 513, in __call__
  return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_manager.py", line 120, in _hookexec
  return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_callers.py", line 139, in _multicall
  raise exception.with_traceback(exception.__traceback__)
File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_callers.py", line 103, in _multicall
  res = hook_impl.function(*args)
        ^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\python.py", line 159, in pytest_pyfunc_call
  result = testfunction(**testargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\Ananya Suresh\Documents\L6_L7_55_car_insurance_premium\LAB7\test_car_insurance_premium_parser.py", line 14, in test_total_claim_amount_no_crash
  def test_total_claim_amount_no_crash(s):
```

```
  File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\hypothesis\core.py", line 2330, in wrapped_test
    raise the_error_hypothesis_found
ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
+---------------- 1 ----------------
| Traceback (most recent call last):
|   File "C:\Users\Ananya Suresh\Documents\L6_L7_55_car_insurance_premium\LAB7\test_car_insurance_premium_parser.py", line 16, in test_total_claim_amount_no_crash
|     total_claim_amount(s)
|   File "C:\Users\Ananya Suresh\Documents\L6_L7_55_car_insurance_premium\LAB7\car_insurance_premium_parser.py", line 16, in total_claim_amount
|     return sum(int(amount) for amount in text.split(';'))
|            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:\Users\Ananya Suresh\Documents\L6_L7_55_car_insurance_premium\LAB7\car_insurance_premium_parser.py", line 16, in <genexpr>
|     return sum(int(amount) for amount in text.split(';'))
|                ^^^^^^^^^^
| ValueError: invalid literal for int() with base 10: ''
|
| During handling of the above exception, another exception occurred:
|
| Traceback (most recent call last):
|   File "C:\Users\Ananya Suresh\Documents\L6_L7_55_car_insurance_premium\LAB7\test_car_insurance_premium_parser.py", line 18, in test_total_claim_amount_no_crash
|     assert False, f"total_claim_amount crashed with: {e!r}"
| AssertionError: total_claim_amount crashed with: ValueError("invalid literal for int() with base 10: ''")
| assert False
| Failing test case: test_total_claim_amount_no_crash(
|     s='',
| )
| Explanation:
|     These lines were always and only run by failing test cases:
|         C:\Users\Ananya Suresh\Documents\L6_L7_55_car_insurance_premium\LAB7\test_car_insurance_premium_parser.py:17
+---------------- 2 ----------------
| Traceback (most recent call last):
|   File "C:\Users\Ananya Suresh\Documents\L6_L7_55_car_insurance_premium\LAB7\test_car_insurance_premium_parser.py", line 16, in test_total_claim_amount_no_crash
|     total_claim_amount(s)
|   File "C:\Users\Ananya Suresh\Documents\L6_L7_55_car_insurance_premium\LAB7\car_insurance_premium_parser.py", line 16, in total_claim_amount
|     return sum(int(amount) for amount in text.split(';'))
|                                          ^^^^^^^^^^
| AttributeError: 'NoneType' object has no attribute 'split'
|
| During handling of the above exception, another exception occurred:
|
| Traceback (most recent call last):
|   File "C:\Users\Ananya Suresh\Documents\L6_L7_55_car_insurance_premium\LAB7\test_car_insurance_premium_parser.py", line 18, in test_total_claim_amount_no_crash
|     assert False, f"total_claim_amount crashed with: {e!r}"
| AssertionError: total_claim_amount crashed with: AttributeError("'NoneType' object has no attribute 'split'")
| assert False
| Failing test case: test_total_claim_amount_no_crash(
|     s=None,
| )
| Explanation:
|     These lines were always and only run by failing test cases:
|         C:\Users\Ananya Suresh\Documents\L6_L7_55_car_insurance_premium\LAB7\test_car_insurance_premium_parser.py:17
```

```
    +----------------------------------
_____ test_driver_age_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\runner.py", line 341, in from_call
  |     result: TResult | None = func()
  |                              ^^^^^^
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\runner.py", line 242, in <lambda>
  |     lambda: runtest_hook(item=item, **kwds), when=when, reraise=reraise
  |             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_hooks.py", line 513, in __call__
  |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_manager.py", line 120, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_callers.py", line 182, in _multicall
  |     return outcome.get_result()
  |            ^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_result.py", line 100, in get_result
  |     raise exc.with_traceback(exc.__traceback__)
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\threadexception.py", line 92, in pytest_runtest_call
  |     yield from thread_exception_runtest_hook()
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\threadexception.py", line 68, in thread_exception_runtest_hook
  |     yield
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\unraisableexception.py", line 95, in pytest_runtest_call
  |     yield from unraisable_exception_runtest_hook()
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\unraisableexception.py", line 70, in unraisable_exception_runtest_hook
  |     yield
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\logging.py", line 846, in pytest_runtest_call
  |     yield from self._runtest_for(item, "call")
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\logging.py", line 829, in _runtest_for
  |     yield
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\capture.py", line 880, in pytest_runtest_call
  |     return (yield)
  |             ^^^^^
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\skipping.py", line 257, in pytest_runtest_call
  |     return (yield)
  |             ^^^^^
  |   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_callers.py", line 103, in _multicall
  |     res = hook_impl.function(*args)
  |           ^^^^^^^^^^^^^^^^^^^^^^^^
```

```
|   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\runner.py", line 174, in pytest_runtest_call
|     item.runtest()
|   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\python.py", line 1627, in runtest
|     self.ihook.pytest_pyfunc_call(pyfuncitem=self)
|   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_hooks.py", line 513, in __call__
|     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
|            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_manager.py", line 120, in _hookexec
|     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
|            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_callers.py", line 139, in _multicall
|     raise exception.with_traceback(exception.__traceback__)
|   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\pluggy\_callers.py", line 103, in _multicall
|     res = hook_impl.function(*args)
|           ^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\_pytest\python.py", line 159, in pytest_pyfunc_call
|     result = testfunction(**testargs)
|              ^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:\Users\Ananya Suresh\Documents\L6_L7_55_car_insurance_premium\LAB7\test_car_insurance_premium_parser.py", line 22, in test_driver_age_no_crash
|     def test_driver_age_no_crash(s):
|                    ^^^
|   File "C:\Users\Ananya Suresh\AppData\Roaming\Python\Python312\site-packages\hypothesis\core.py", line 2330, in wrapped_test
|     raise the_error_hypothesis_found
| ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
+-+---------------- 1 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\Ananya Suresh\Documents\L6_L7_55_car_insurance_premium\LAB7\test_car_insurance_premium_parser.py", line 24, in test_driver_age_no_crash
  |     driver_age(s)
  |   File "C:\Users\Ananya Suresh\Documents\L6_L7_55_car_insurance_premium\LAB7\car_insurance_premium_parser.py", line 23, in driver_age
  |     return current_year - int(dob.split('/')[2])
  |                           ~~~~~~~~~~~~~~~~~~^^^
  | IndexError: list index out of range
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "C:\Users\Ananya Suresh\Documents\L6_L7_55_car_insurance_premium\LAB7\test_car_insurance_premium_parser.py", line 26, in test_driver_age_no_crash
  |     assert False, f"driver_age crashed with: {e!r}"
  | AssertionError: driver_age crashed with: IndexError('list index out of range')
  | assert False
  | Failing test case: test_driver_age_no_crash(
  |     s='',  # or any other generated value
  | )
  +---------------- 2 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\Ananya Suresh\Documents\L6_L7_55_car_insurance_premium\LAB7\test_car_insurance_premium_parser.py", line 24, in test_driver_age_no_crash
  |     driver_age(s)
  |   File "C:\Users\Ananya Suresh\Documents\L6_L7_55_car_insurance_premium\LAB7\car_insurance_premium_parser.py", line 23, in driver_age
  |     return current_year - int(dob.split('/')[2])
  |                           ^^^^^^^^^
  | AttributeError: 'NoneType' object has no attribute 'split'
```

```
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "C:\Users\Ananya Suresh\Documents\L6_L7_55_car_insurance_premium\LAB7\test_car_insurance_premium_parser.py", line 26, in test_driver_age_no_crash
  |     assert False, f"driver_age crashed with: {e!r}"
  | AssertionError: driver_age crashed with: AttributeError("'NoneType' object has no attribute 'split'")
  | assert False
  | Failing test case: test_driver_age_no_crash(
  |     s=None,  # or any other generated value
  | )
  +-----------------------------------
=========================================================== short test summary info ============================================================
FAILED test_car_insurance_premium_parser.py::test_normalize_registration_no_crash - AssertionError: normalize_registration crashed with: AttributeError("'NoneType' object has no attribute 'strip'")
FAILED test_car_insurance_premium_parser.py::test_total_claim_amount_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_car_insurance_premium_parser.py::test_driver_age_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
============================================================ 3 failed in 6.36s =============================================================
```

Fixed module run

`python car_insurance_premium_parser.py` on the fixed module, with the same inputs as SS1. The empty claim entry is skipped (total 12800) and the invalid date of birth is reported instead of crashing.

```
PS C:\Users\Ananya Suresh\Documents\SE_LAB_PES2UG24CS059\lab7> python car_insurance_premium_parser.py
==== Running normalize_registration ====
Enter a registration number (e.g. ka 01 ab 1234): ka 01 gh 3456
Normalized registration: KA01GH3456

==== Running total_claim_amount ====
Enter past claim amounts separated by ; (e.g. 12000;4500;800): 12000;;800
Total claimed: 12800

==== Running driver_age ====
Enter the driver's date of birth as DD/MM/YYYY (e.g. 14/08/1999): 1999
Driver age in 2026: Invalid date of birth
PS C:\Users\Ananya Suresh\Documents\SE_LAB_PES2UG24CS059\lab7>
```

Tests passing on the fixed module

`python -m pytest test_car_insurance_premium_parser.py` on the fixed module. All 3 tests pass.

```
PS C:\Users\Ananya Suresh\Documents\SE_LAB_PES2UG24CS059\lab7> python -m pytest test_car_insurance_premium_parser.py
============================= test session starts =============================
platform win32 -- Python 3.12.6, pytest-8.3.4, pluggy-1.5.0
rootdir: C:\Users\Ananya Suresh\Documents\SE_LAB_PES2UG24CS059\lab7
plugins: anyio-4.15.1, hypothesis-6.170.14
collected 3 items

test_car_insurance_premium_parser.py ...                                 [100%]

============================== 3 passed in 0.78s ==============================
PS C:\Users\Ananya Suresh\Documents\SE_LAB_PES2UG24CS059\lab7>
```

Bugs found and fixes

All three functions assumed well-formed input. Hypothesis found `None` crashing every function, plus malformed strings in two of them.

Function	Bug	Crashing input	Fix
normalize_registration	Assumes input is a string	`None` (AttributeError)	Type check, return an empty string for non-strings
total_claim_amount	Assumes every piece is a whole number	`""`, `"12000;;800"`, `"abc"` (ValueError); `None` (AttributeError)	Skip invalid or negative pieces, return 0 for non-strings
driver_age	Assumes exactly three parts and a valid year	`"1999"` (IndexError); `"a/b/c"` (ValueError); `None` (AttributeError)	Check part count, validate with `date()`, return `None` for invalid or future dates

Reflection questions

Q1. What bugs did fuzzing find that you might have missed with manual testing?

I only tried a few inputs by hand, such as 12000;;800 and 1999. Hypothesis found more, including None, which crashed all three functions with AttributeError. It also tried empty strings, non-numeric text like "abc", and dates that are malformed or out of range, which caused

ValueError and IndexError. It then shrank each failure to a minimal input, so it was clear what to fix.

Q2. How did you fix the bugs, and why is this defensive programming?
Each function now checks its input before using it, instead of assuming it is well formed. normalize_registration returns an empty string for non-strings. total_claim_amount skips empty, invalid or negative entries and returns 0 for non-strings. driver_age checks that there are three parts, validates the date with datetime.date(), and returns None for invalid or future dates. The result is that bad input is handled gracefully instead of crashing the program.

Q3. What are the limitations of this kind of fuzzing, and how could you improve the tests?
My tests only check that the functions don't crash. They don't check that the answers are correct, and a function that returns 0 for every input would pass. I could add property checks, such as the result of normalize_registration never containing spaces or lowercase letters, total_claim_amount never being negative, and driver_age never being negative. I could also use more targeted strategies, like generated dates and lists of numbers, instead of random text only.